

1) Answer any **two** of the following:

- a) Create a generic method called getMax that takes an array of comparable elements and returns the maximum element from the array. Implement a Java program that demonstrates the usage of this method with different types of arrays, such as Integer[], String[], and Double[]. [5]
- b) Explain why a *List<Integer>* is not considered a subtype of *List<Number>*, even though Integer is a subtype of Number. Provide examples to illustrate the concept and explain how bounded wildcards can be used to handle such scenarios. [5]
- c) Explain the limitations and restrictions of generics in Java considering the situation of i) Creating Arrays of Parameterized Types and ii) Using the instanceof Operator with Generic Types. [5]

2) Answer **a)** and any **one** from remaining:

- a) Discuss strength and weakness of the sequence and collaboration. [5]
- b) Consider a Dice Game, where a player picks up and rolls two dices. If the dice face value is total seven, he/she wins; otherwise losses.

- i. Design and explain Class diagram, also identify classes and their relationships.
- ii. Design Sequence diagram for the above.
- iii. Also design the Partial Model for the dice game.

c) Consider a university system that consists of multiple schools, departments, students, courses, and instructors. [10]
Based on this scenario, design a class diagram to represent the relationships and attributes of these entities in the system.
Use appropriate UML notations, also keeping in mind the following while designing the class diagram:

- (i) Correctly identify and represent the entities.
- (ii) Establish the relationships between entities.
- (iii) Including relevant attributes for each entity.

3) Answer the following:

- a) What are the SOLID principles of object-oriented design?
- b) **Explain** Single-Responsibility Principle.

[10]

[1+4]